1. P3197 越狱

题目描述

监狱有 n 个房间，每个房间关押一个犯人，有 m 种宗教，每个犯人会信仰其中一种。如果相邻房间的犯人的宗教相同，就可能发生越狱，求有多少种状态可能发生越狱。

答案对 $100,003$ 取模。

输入格式

输入只有一行两个整数，分别代表宗教数 m 和房间数 n。

输出格式

输出一行一个整数代表答案。

输入输出样例 #1

输入 #1

```
2 3
```

输出 #1

```
6
```

说明/提示

样例输入输出 1 解释

状态编号	**1** 号房间	**2** 号房间	**3** 号房间
1	信仰 1	信仰 1	信仰 1
2	信仰 1	信仰 1	信仰 2
3	信仰 1	信仰 2	信仰 2
4	信仰 2	信仰 1	信仰 1
5	信仰 2	信仰 2	信仰 2
6	信仰 2	信仰 2	信仰 1

数据规模与约定

对于 100% 的数据，保证 $1 \le m \le 10^8$，$1 \le n \le 10^{12}$。

2. Halloween Treats

Every year there is the same problem at Halloween: Each neighbour is only willing to give a certain total number of sweets on that day, no matter how many children call on him, so it may happen that a child will get nothing if it is too late. To avoid conflicts, the children have decided they will put all sweets together and then divide them evenly among themselves. From last year's experience of Halloween they know how many sweets they get from each neighbour. Since they care more about justice than about the number of sweets they get, they want to select a subset of the neighbours to visit, so that in sharing every child receives the same number of sweets. They will not be satisfied if they have any sweets left which cannot be divided.

Your job is to help the children and present a solution.

Input

The input contains several test cases.
The first line of each test case contains two integers **c** and **n** ($1 \le c \le n \le 100000$), the number of children and the number of neighbours, respectively. The next line contains **n** space separated integers a_1 , ... , a_n ($1 \le a_i \le 100000$), where a_i represents the number of sweets the children get if they visit neighbour ***i***.

The last test case is followed by two zeros.

Output

For each test case output one line with the indices of the neighbours the children should select (here, index ***i*** corresponds to neighbour ***i*** who gives a total number of a_i sweets). If there is no solution where each child gets at least one sweet print "no sweets" instead. Note that if there are several solutions where each child gets at least one sweet, you may print any of them.

Sample

Input 复制	Output 复制
4 5 1 2 3 7 5 3 6 7 11 2 5 13 17 0 0	3 5 2 3 4

3. P1287 盒子与球

题目描述

现有 r 个互不相同的盒子和 n 个互不相同的球，要将这 n 个球放入 r 个盒子中，且不允许有空盒子。请求出有多少种不同的放法。

两种放法不同当且仅当存在一个球使得该球在两种放法中放入了不同的盒子。

输入格式

输入只有一行两个整数，分别代表 n 和 r。

输出格式

输出一行一个整数代表答案。

输入输出样例 #1

输入 #1

```
3 2
```

输出 #1

```
6

```

说明/提示

样例输入输出 1 解释

有两个盒子（编号为 $1,2$）和三个球（编号为 $1,2,3$），共有六种方案，分别如下：

盒子编号	方案 1	方案 2	方案 3	方案 4	方案 5	方案 6
盒子 1	小球 1	小球 2	小球 3	小球 $2,3$	小球 $1,3$	小球 $1,2$
盒子 2	小球 $2,3$	小球 $1,3$	小球 $1,2$	小球 1	小球 2	小球 3

数据规模与约定

对于 100% 的数据，保证 $0 \le r \le n \le 10$，且答案小于 2^{31}。